

TURAN ORUJLU

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EDUCATION

International Max Planck Research School for Intelligent Systems

Dec 2022 – Present

Doctor of Philosophy Candidate

Supervisors: Charley M. Wu and Martin V. Butz

University of Tübingen, Tübingen, Germany

Nov 2020 – Sep 2022

Master of Science, Neural Information Processing

University of Osnabrück, Osnabrück, Germany

Apr 2016 – Nov 2020

Bachelor of Science, Cognitive Science

University of Pennsylvania, Philadelphia, PA, USA

Sep 2010 – Dec 2013

Bachelor of Arts, Mathematical Economics

Minor: Chemistry

RESEARCH INTERESTS

Causality · Model-based Reinforcement Learning · Intuitive Physics · Mechanistic Interpretability

PUBLICATIONS

[1] Partially Observed Structural Causal Models

Turan Orujlu, Jordan Matelsky, Martin V. Butz, Charley M. Wu, Konrad P. Kording

UAI 2026 · Amsterdam, Netherlands · Aug 2026

Poster

[2] Causal Process Models: Reframing Causal Graph Discovery as a Reinforcement Learning Problem

Turan Orujlu, Christian Gumbsch, Martin V. Butz, Charley M. Wu

CLear 2026 · Boston, MA, USA · Apr 2026

Poster

[3] Causal Substrates: A Set-Theoretic Foundation of Causal Inference

Turan Orujlu, Moritz Willig, Matej Zečević

Causality for Decision Making Workshop, UAI 2026 · Amsterdam, Netherlands · Aug 2026

Workshop

[4] Reframing Attention as a Reinforcement Learning Problem for Causal Discovery

Turan Orujlu, Christian Gumbsch, Martin V. Butz, Charley M. Wu

Causal RL Workshop, RLC 2025 · Edmonton, AB, Canada · Aug 2025

Workshop Oral

TALKS

Partially Observed Structural Causal Models

– Shpitser Lab (PI: Ilya Shpitser), Johns Hopkins University

Apr 2026

Reframing Attention as a Reinforcement Learning Problem for Causal Discovery

– Computational Machinery of Cognition Lab (PI: Shervin Safavi), TU Dresden

Dec 2025

– Causal RL Workshop, RL Conference

Aug 2025

SUPERVISION

Trong Vu Le

2025

Master Rotation · University of Tübingen

The role of positional embeddings in transformer-based transition models for representation learning in compositional environments

Marcel De Sutter

2023

Master Thesis · University of Tübingen

Physical and psychological reasoning in artificial cognitive systems: a benchmark study of Loci's object vs. agent recognition

RESEARCH EXPERIENCE

Tübingen AI Center, University of Tübingen

Dec 2022 – Present

Research Assistant

- Conducting research towards completion of the doctoral degree requirements.

University of Pennsylvania

Oct 2025 – Dec 2025

Research Scholar, Kording Lab (PI: Konrad Kording)

- Research visit.

Max Planck Institute for Biological Cybernetics

Nov 2021 – Nov 2022

Research Assistant, Computational Neuroscience Department (PI: Peter Dayan)

- Master thesis project.
- Lab rotation.

Johns Hopkins University

August 2019 – Dec 2019

Research Scholar, Vision Lab (PI: René Vidal)

- Research visit.

TEACHING EXPERIENCE

Tübingen AI Center, University of Tübingen

Apr 2023 – Jun 2023

Nov 2024 – Feb 2025

Teaching Assistant, Human and Machine Cognition Lab

- Corrected weekly homework for the *General Principles of Human and Machine Learning* lecture.
- Hosted weekly tutorials for students of the *General Principles of Human and Machine Learning* lecture.

Institute of Cognitive Science, University of Osnabrück

Oct 2017 – Feb 2018

Oct 2018 – Feb 2019

Teaching Assistant, Neuroinformatics Research Group

- Corrected weekly homework for the *Neuroinformatics* lecture.
- Hosted weekly Q & A sessions for students of the *Neuroinformatics* course.
- Corrected the final exam for the *Neuroinformatics* course.

Institute of Mathematics, University of Osnabrück

Oct 2018 – Feb 2019

Teaching Assistant, Stochastics Research Group

- Corrected weekly homework for the *Probability* lecture.
- Hosted weekly tutorials for students of the *Probability* course.
- Corrected the final exam for the *Probability* course.

SERVICE

- Organized the RLDM talk series at the MPI for Biological Cybernetics.
- Reviewer for CogSci 2026 and the NeurIPS 2026 Position Paper Track.

STIPENDS AND GRANTS

- Tübingen ML Cluster's Research Visit Abroad Program grant for the research stay at the University of Pennsylvania
- Stipend covering the Master phase of the 5-year track of the IMPRS for the Mechanisms of Mental Function and Dysfunction.
- DAAD-PROMOS travel grant for the research stay at Johns Hopkins University.
- DAAD RISE Worldwide Scholarship. Aug – Oct 2019
- Germany Scholarship (Deutschlandstipendium). 2018/19

SUMMER SCHOOLS

- School on Analytical Connectionism (UCL, 2023)
- Machine Learning Summer School (Columbia, 2026)